



Efficacy of manual therapy for sacroiliac joint pain syndrome: a systematic review and meta-analysis of randomized controlled trials

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ABSTRACT

Introduction: This study examined the efficacy of manual therapy for pain and disability measures in adults with sacroiliac joint pain syndrome (SIJPS).

Methods: We searched six databases, including gray literature, on 24 October 2023, for randomized controlled trials (RCTs) examining sacroiliac joint (SIJ) manual therapy outcomes via pain or disability in adults with SIJPS. We evaluated quality via the Physiotherapy Evidence Database scale and certainty via Grading of Recommendations, Assessment, Development, and Evaluation (GRADE). Standardized mean differences (SMDs) in post-treatment pain and disability scores were pooled using random-effects models in meta-regressions.

Results: We included 16 RCTs (421 adults; mean age = 37.7 years), with 11 RCTs being meta-analyzed. Compared to non-manual physiotherapy (i.e. exercise ± passive modalities; 10 RCTs) or sham (1 RCT) interventions, SIJ manual therapy did not significantly reduce pain (SMD: -0.88; 95% CI: -1.84; 0.08, $p = 0.0686$) yet had a statistically significant moderate effect in reducing disability (SMD: -0.67; 95% CI: -1.32; -0.03, $p = 0.0418$). The superiority of individual manual therapies was unclear due to low sample size, wide confidence intervals for effect estimates, and inability to meta-analyze five RCTs with a unique head-to-head design. RCTs were of 'good' (56%) or 'fair' (44%) quality, and heterogeneity was high. Certainty was very low for pain and low for disability outcomes.

Conclusion: SIJ manual therapy appears efficacious for improving disability in adults with SIJPS, while its efficacy for pain is uncertain. It is unclear which specific manual therapy techniques may be more efficacious. These findings should be interpreted cautiously until further high-quality RCTs are available examining manual therapy against control groups such as exercise.

Registration: PROSPERO (CRD42023394326).

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Introduction

Sacroiliac joint pain syndrome (SIJPS) is increasingly recognized as an important cause of low back pain (LBP), with 10–38% of the cases of adult LBP being related to SIJPS [1]. SIJPS can be defined as pain and dysfunction of the sacroiliac joint (SIJ) and surrounding soft tissue structures, which can cause local LBP or referred pain in the associated lower extremity [1]. Unfortunately, there is no consensus on the most appropriate treatment for SIJPS [2]. Treatments for SIJPS range from conservative options to surgical management, with conservative treatment options frequently being used for first-line management of SIJPS [2–4]. Conservative interventions may include injections, radiofrequency ablations, exercises, electrotherapeutic modalities, and manual therapy [2–4].

Manual therapy interventions for SIJPS include high velocity, low amplitude (HVLA) thrust manipulation, low-velocity mobilization, soft tissue techniques, and muscle energy techniques (MET) [2]. Although manipulation is supported by evidence for the treatment of LBP, there are limited data regarding its utility for treating SIJPS [2]. In addition, it remains unclear if the use of manual therapy provides superior outcomes in comparison to other conservative interventions, such as exercise [2,5,6]. While previous systematic reviews have examined the efficacy of exercise and injections for SIJPS [2,7,8], to our knowledge, none have focused on the efficacy of manual therapy.

Considering the lack of previous evidence synthesis on this topic, the primary objective of this systematic review with meta-analysis was to

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examine the efficacy of manual therapy for SIJPS in adults with respect to pain and disability.

Methods

Protocol and registration

The current study is a systematic review and meta-analysis examining the efficacy of SIJ manual therapy on pain and disability outcomes in adult patients with SIJD. The review protocol was registered a priori with PROSPERO (CRD42023394326). We had multiple deviations from the original protocol: (1) We restricted included studies to randomized controlled trials (RCTs) only, as we identified a sufficient number of such studies, (2) omitted lumbar spine range of motion as a study outcome given the lack of trials examining this outcome, (3) added more databases to our search, (4) incorporated meta-regression to compare efficacy and explore heterogeneity of different manual therapies, and (5) added another author (RT). Study reporting adheres to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 statement [9].

Eligibility criteria

Inclusion criteria were adults (age ≥ 18), with SIJPS supported by at least one positive SIJ pain provocation test, use of manual techniques for treatment of SIJPS, RCT study design, follow-up of at least 1 week, and outcomes related to pain (i.e. numeric pain rating scale [NPRS] or visual analog scale [VAS]) and patient-reported measures of disability or function (i.e. Oswestry Disability Index, [ODI], or Modified ODI [MODI]). Exclusion criteria were articles describing patients without SIJPS, study designs other than RCTs, no manual techniques used as an intervention, follow-up less than 1 week, and not containing outcomes related to pain or disability.

Information sources

We searched PubMed, the Cumulative Index to Nursing and Allied Health Literature (CINAHL), Scopus, Web of Science, Google Scholar, and Ovid MEDLINE® on 24 October 2023, from inception to the date of query. Gray literature was included via screening the first 50 search results from Google Scholar (queried and exported via Harzing's Publish or Perish 7 (Tarma Software Research Ltd.®)) as well as through citation tracking of included articles and asking coauthors to submit relevant articles from their personal collection if missed by the main searches.

Search strategy

We require articles to include terms related to the sacroiliac joint, terms for manual therapy interventions, and methodology terms indicating RCT design. We required sacroiliac joint terms including 'sacroiliac', 'SI joint', or 'SIJ'. For interventions, we required terms such as 'manipulation', 'mobilization', 'myofascial release', or 'chiropractic'. For methodology, we required mentions of 'randomized', 'randomized', or 'RCT'. See the full search for PubMed and other databases summarized in Supplemental File 1.

Selection process

Duplicates were manually removed prior to article screening. Article selection was conducted using Rayyan software [10]. Articles were then screened by title and abstract. Finally, articles were screened by full text for inclusion in this study. Two authors independently conducted each phase of article screening (AB and RT) and disagreements were resolved by discussion.

Data collection process

Data extraction was performed initially by a single author (RT) and then verified by a second author (AB) with any disagreements resolved by discussion. Extracted data included author surname, year of publication, number of positive SIJ pain provocation tests, number of patients per study and comparator group, age of patients, percentage of female patients, types of pertinent outcome assessments for our study (i.e. VAS, NPRS, ODI, and MODI), post-treatment pain score and standard deviation (SD), post-treatment ODI/MODI score and SD, weeks of follow-up duration, and type of interventions per study group.

Data items

For this study, SIJPS was defined as a subset of LBP that arises from the SIJ which can present with unilateral LBP and buttock and/or lower extremity pain, and is reproduced by SIJ provocation maneuvers [1,3]. Non-manual physiotherapy was defined as exercise or passive modalities (ultrasound therapy or electrotherapy) [2]. The broadest category of treatment used for our primary objective, SIJ manual therapy, was defined as any myofascial release, SIJ low-grade mobilization, MET, or SIJ HVLA manipulation directed to the SIJ [2]. SIJ HVLA manipulation was defined as the use of a manual thrust or impulse performed in the side-lying, prone, or supine position that targets the SIJ to improve mobility, decrease pain, and/or improve function [2,5]. SIJ METs were defined as any of a group of brief contract-relax cycles performed by the patient

against resistance from the clinician to improve SIJ asymmetries, pain, and/or function [11–13]. SIJ myofascial release was defined as any soft tissue or massage technique provided to the tissues overlying or surrounding the SIJ [14]. SIJ HVLA manipulation, MET, and myofascial release were each considered subsets of SIJ manual therapy. SIJ provocation maneuvers included the thigh thrust test, sacral thrust test, Yeoman's test, distraction test, compression test, Gaenslen's test, and the Flexion, ABduction and External Rotation (FABER) test [15–17].

Bias and quality assessment

The Physiotherapy Evidence Database (PEDro) scale for RCTs was used to evaluate the bias and quality of all included studies and was reported using a 0–10 scale [18]. PEDro scores were interpreted as 0–3 = poor, 4–5 = fair, 6–8 = good, and 9–10 = excellent methodological quality [19]. Scores were derived by a single author (RT) and verified by another (AB) with disagreements resolved by discussion.

Synthesis methods

We conducted continuous model meta-analyses to synthesize the effects of manual therapy on post-treatment pain and disability scores in SIJPS. We included RCTs in meta-analysis when there were at least three methodologically similar studies, for example comparing manual therapy vs. control. For head-to-head RCTs comparing different manual therapy techniques, we planned to conduct separate meta-analyses if at least three RCTs examined sufficiently similar manual therapy technique comparisons; otherwise, these were summarized narratively. We use standardized mean difference (SMD) as a generalizable effect size metric to account for slightly different measurement scales (i.e. ODI vs. MODI; VAS vs. NPRS) [20] across studies. Our use of between-group comparisons of post-treatment scores was intended to control for variable follow-up duration, natural recovery, regression to the mean, and effects of concurrent interventions like exercise that would be shared by both groups [21]. Our strategy was therefore intended to provide a more direct measure of the efficacy of manual therapy, even when both groups received a usual care intervention such as exercise.

To calculate SMDs, we used post-treatment means, SDs, and group sample sizes for pain and disability outcomes from the longest follow-up assessment available. The SMDs were calculated using Cohen's *d*, with values of 0.2, 0.5, and 0.8 reflecting trivial, moderate, and large effect sizes, respectively [22]. Analyses were performed separately for pain and disability outcomes. Meta-regression analyses were conducted to examine the effects of specific manual therapy techniques or combinations. Between-study heterogeneity

was assessed using the *I* [2] statistic, interpreted as <25% = low, 25–50% = moderate, and >50% = high heterogeneity [23]. All meta-analyses were conducted using Meta-Mar software [24] (version 3.5.118).

We used a random effects model based on anticipated heterogeneity related to included trial populations, protocols, follow-up durations, and manual therapy techniques. The random effects model accounts for both within- and between-study variance, allowing for differences in true effects across studies. This approach is preferable for meta-analyses with heterogeneity [25]. Findings with a *p*-value < 0.05 were considered statistically significant.

We used R and R studio (version 4.2.2, Vienna, AT) [26] to calculate summary measures for study characteristics. We calculated the weighted mean age from studies reporting means and SDs, accounting for varying sample sizes within and across studies. We imputed missing SD from one study using pain and disability score, sample size, and *p*-value [27]. Kappa values for screening were calculated using the formula for inter-rater agreement, with values reflecting agreement strength: <0.20 is 'poor', '0.21–0.40' is fair, 0.41–0.60 is 'moderate', 0.61–0.80 is 'good', and '0.81–1.00' is very good [28].

Reporting bias assessment

Potential publication bias was evaluated by visual inspection of funnel plots and Egger's regression test when the analysis included at least 10 studies [29,30].

Certainty assessment

We used the Grading of Recommendations, Assessment, Development, and Evaluation (GRADE) approach to assess the certainty of evidence for each analysis [31]. GRADE scores were derived by a single author (RT) and verified by a second (AB) with discrepancies resolved via discussion. The certainty of evidence was rated as high, moderate, low, or very low, based on the study design, risk of bias (i.e. PEDro score), inconsistency (i.e. heterogeneity), indirectness, and publication bias of the evidence.

Results

Study selection

Of 281 articles retrieved from the database search, 157 were screened, and 29 were assessed for eligibility. Two additional articles were obtained from citation tracking and from the authors' personal collections [11,32]. For database title/abstract screening, reviewers had 96% agreement, yielding a kappa value of 0.85 (very good). For database full-text screening, reviewers had 76% agreement, yielding a kappa value of 0.48 (moderate).

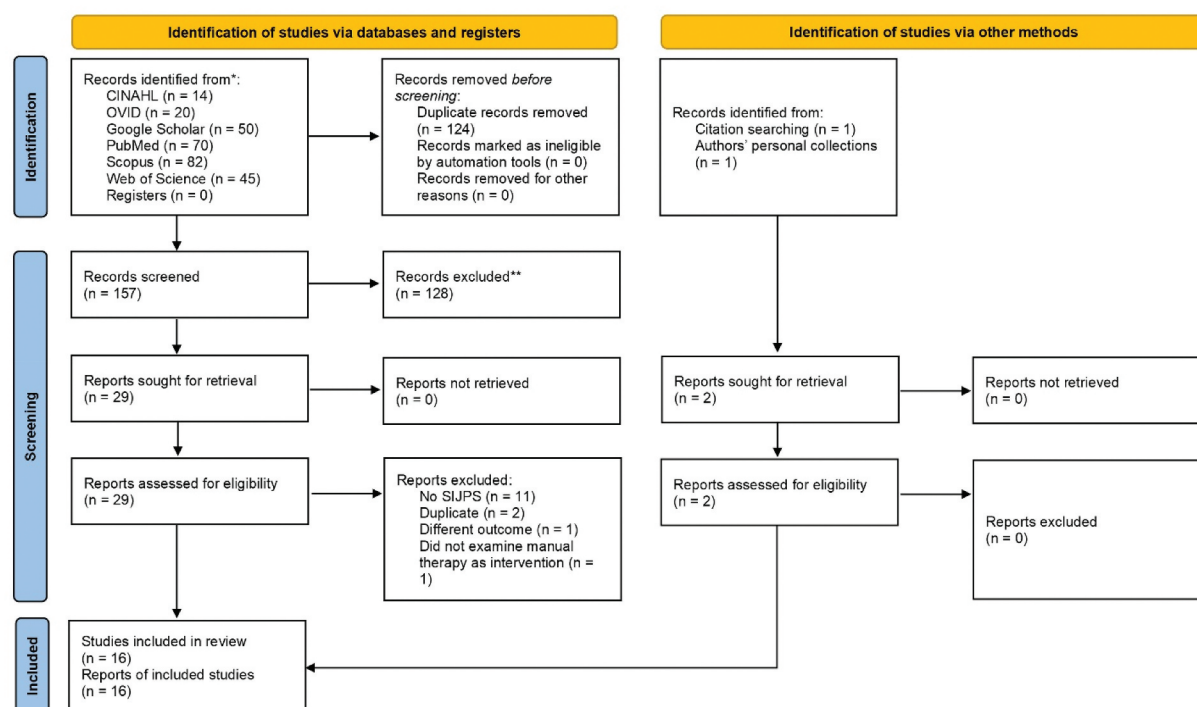


Figure 1. Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow diagram outlining the initial search to final article inclusion. Abbreviations: sacroiliac joint pain syndrome (SIJPS).

All disagreements were resolved via discussion. Sixteen studies met our final selection criteria (Figure 1) [11–14,27,32–42]. We excluded 14 articles at full-text screening for the following reasons: did not require SIJPS ($n = 11$) [39,43–52], two of which were a duplicate ($n = 2$) [49,50], did not include standardized pain or disability measures ($n = 1$) [53], or did not examine SIJ manual therapy as an intervention ($n = 1$) [54].

Study characteristics

The 16 trials included a total of 776 adults with SIJPS (Table 1). The weighted mean age of the included patients was 37.7 (SD = 8.2) years. The mean follow-up time for the included patients was 6 (SD = 6) weeks (median = 4; Q1:Q3 = 2:9), while only five trials had a follow-up longer than 4 weeks. After accounting for drop-outs, a total of 458 out of 664 patients received manual therapies for SIJPS (69%). Of these patients, the majority received SIJ HVLA thrust manipulation ($n = 259$, 57%), followed by mobilization ($n = 134$, 29%), MET ($n = 134$, 29%), and myofascial release ($n = 32$, 7%). All trials required pain in the SIJ region and at least three positive SIJ provocation tests for a diagnosis of SIJPS [15–17].

Eleven RCTs were able to be included in meta-analyses due to having a similar methodological design of comparator groups including manual therapy versus a control group [11,13,27,32,33,35,37,39–42]. In 10 of these RCTs, groups compared manual versus non-manual physiotherapy: (1) manual therapy plus exercise versus (2) exercise alone (9 RCTs) [13,27,32,33,35,37,39,40,42] or (1)

manual therapy plus exercise and ultrasound therapy versus (2) exercise and ultrasound therapy alone (1 RCT) [11]. One RCT included a control group receiving a sham intervention in which patients approximated a position used in MET, yet there was no contract-relax intervention [41]. While one meta-analyzed RCT included a third comparator group of injection [42], this specific group could not be included due to it being the sole study using injection as a comparator. One of the meta-analyzed RCTs had three eligible comparator groups (1. HVLA manipulation + exercises; 2. MET + exercises; 3. exercises) and both relevant comparisons were included in the meta-analysis (i.e. group 1 vs. 3; group 2 vs. 3) [32].

Risk of bias in studies

Per PEDro scoring, the methodological quality was deemed ‘good’ in nine RCTs (i.e. 56%) [12–14,34,36,37,40–42], and ‘fair’ in seven (44%) [11,27,32,33,35,38,39] (Table 2). No RCT had ‘excellent’ or ‘poor’ methodological quality.

Results of syntheses

SIJ manual therapy for pain

Our meta-regression including 11 trials with 12 intervention-control comparisons indicated a plausible effect of manual therapy on reducing SIJP (Figure 2), indicated by a large negative SMD of -0.88 (95%-CI: -1.84 ; 0.08 , $p = 0.0686$). However, as the p -value exceeded 0.05, and the confidence interval included zero, there was a lack of

Table 1. Study characteristics. Abbreviations: high-velocity, low-amplitude (HVLA); mechanical diagnosis and therapy; numeric pain rating scale (NPRS); sacroiliac joint (SIJ); sacroiliac joint pain syndrome (SIJPS); ultrasound (US); visual analogue scale (VAS). Reporting of patient age varied per study and in some instances, we were only able to report the minimum and maximum values rather than individual group means or vice versa.

Author (Year)	Age, mean (SD) and/or min-max	Female (%)	Interventions & group sizes after dropouts (n)	Initial sample size	Outcomes	Total weeks of follow-up	Key outcome(s)
Castro-Sánchez (2021)	1. 53 (8) 2. 50 (5)	60	1. SIJ HVLA manipulation (32) 2. Myofascial release (32)	64	NPRS, ODI	12	No significant between-group difference
Dogan (2021)	1. 35.1 (13.9) 2. 39.0 (11.3)	68	1. Home exercises (32) 2. SIJ mobilization + home exercises (32)	64	VAS	4	No significant between-group difference
Fatima (2021)	1. 30.47 (8.16) 2. 31.93 (6.83)	60	1. SIJ HVLA manipulation (15) 2. MET (15)	30	NPRS, MODI	2	No significant between-group difference
Hatik (2022)	1. 39.47 (9.92) 2. 35.07 (10.06)	64	1. HVLA manipulation + exercises (15) 2. MET + exercises (15) 3. Home exercises (15)	45	ODI	4	CM more efficacious than MET + exercise and exercise alone
Hussein (2022)	1. 34.35 (10.76) 2. 34.35 (9.69)	57	1. Strain-counterstrain (28) 2. MET (28)	56	ODI	4	MET more efficacious than strain-counterstrain
Javadov (2021)	1. 32.8 (6.7) 2. 36.2 (9.1) 3. 36.9 (9.6)	100	1. SIJ HVLA manipulation and SIJ exercise (23) 2. SIJ HVLA manipulation and lumbar exercise (23) 3. Lumbar exercise (23)	69	VAS, ODI	12	Manual therapy and exercise more efficacious than exercise alone
Kamali (2012)	1. 24.75 (3.53) 2. 24.25 (3.13)	100	1. SIJ HVLA manipulation (16) 2. SIJ & lumbar HVLA manipulation (16)	32	VAS, ODI	4	SIJ and lumbar manipulation was more efficacious for disability than SIJ manipulation
Kamali (2019)	1. 40.2 (14.26) 2. 42.07 (14.03)	70	1. Stabilization exercise (15) 2. SIJ HVLA manipulation (15)	40	VAS, ODI	1.4 2.2	No significant between-group difference
Mathew (2015)	18–30	Unclear	1. US and exercise (15) 2. MET and SIJ mobilization (low velocity) with US and exercise (15)	30	VAS, MODI	4	MET and mobilization more efficacious than US and exercise
Nejati (2019)	46.8 (23–60)	58	1. Exercises (17) 2. SIJ HVLA manipulation & mobilization (17) 3. Exercises, HVLA manipulation, and mobilization (17)	51	VAS, ODI	24	No significant between-group differences
Sanika (2021)	1. 38.83 (11.4) 2. 34.96 (9.5)	67	1. MET and gluteus maximus activation (24) 3. MET and flexion bias exercises (24)	48	VAS, ODI	4	No significant between-group differences
Sivakumar (2022)	24–47	Unclear	1. Mulligan mobilization and motor control exercises (30) 2. Motor control exercises (33) 3. Aerobic exercises (32)	95	NPRS, ODI	6	Mulligan mobilization and exercise more efficacious than exercise for pain and disability
Srivastava (2020)	1. 31.80 (8.62) 2. 41.20 (12.07)	60	1. MET (10) 2. Mechanical diagnosis and therapy (10)	20	VAS	1	No significant between-group differences
Tahir (2022)	1. 40.23 (6.34) 2. 38.76 (5.80)	44	1. Stabilization exercises (17) 2. MET (17)	34	NPRS, MODI	4	MET more efficacious for pain and disability
Vaseghnia (2019)	29.6 (5.30)	100	1. MET (25) 2. Sham (25)	50	VAS, ODI	1	MET more effective for pain and disability
Visser (2013)	46.2 (13.9)	73	1. Exercise (5) 2. SIJ HVLA manipulation (12) 3. Injection (13)	51	VAS	12	SIJ HVLA manipulation more efficacious than exercise

Table 2. Bias and quality assessment grading via the PEDro scale for the included articles. Each category was graded with a zero (absent) or one (present). Articles also received a total score (ranging 0–10 points).

Author (Year)	PEDro Score	1: Eligibility criteria were specified	2: Random allocation	3: Allocation was concealed	4: Groups similar at baseline	5: Subject blinding	6: Therapist blinding	7: Assessor blinding	8: Less than 15% dropouts	9: Intention-to-treat analysis	10: Between-group statistical comparisons	11: Point measures and variability data
Castro-Sánchez (2021)	7	Yes	1	1	1	0	0	0	1	1	1	1
Dogan (2021)	4	Yes	1	0	1	0	0	0	0	0	1	1
Fatima (2021)	8	Yes	1	1	1	1	0	1	1	0	1	1
Hatik (2022)	4	Yes	1	0	1	0	0	0	0	0	1	1
Hussein (2022)	6	Yes	1	1	1	0	0	0	1	0	1	1
Javadov (2021)	5	Yes	1	0	0	0	0	0	1	1	1	1
Kamali (2012)	6	Yes	1	0	1	0	0	0	1	1	1	1
Kamali (2019)	6	Yes	1	1	1	0	0	1	0	0	1	1
Mathew (2015)	5	Yes	1	0	1	0	0	0	1	0	1	1
Nejati (2019)	5	Yes	1	0	1	0	0	1	1	0	1	0
Sanika (2021)	5	Yes	1	0	1	0	0	0	1	0	1	1
Sivakumar (2022)	5	Yes	1	0	1	0	1	0	0	0	1	1
Srivastava (2020)	8	Yes	1	1	1	0	0	1	1	1	1	1
Tahir (2022)	7	Yes	1	1	1	1	0	0	1	0	1	1
Vaseghnia (2019)	6	Yes	1	0	1	0	0	1	1	0	1	1
Visser (2013)	6	Yes	1	0	1	0	0	1	0	1	1	1

clear statistical significance. Significant differences were evident between subgroups ($p < 0.01$), suggesting that the efficacy of manual therapy may depend on the technique(s) used. Direct subgroup comparison was limited due to only two groups (SMT and MET) having more than one included study. However, the effect estimates for these subgroups (SMT and MET) had overlapping confidence intervals, suggesting a plausibly similar efficacy.

Significant heterogeneity was present across the included studies ($I^2 = 91\%$, $p < 0.01$). The type of manual therapy (e.g. HVLA manipulation or MET) did not explain this heterogeneity based on meta-regression results (all p values > 0.05 ; Figure 2). Therefore, the high level of heterogeneity may be attributed to methodological differences in study populations, interventions, and follow-up durations. Visual inspection of a contour-enhanced funnel plot showed a generally symmetrical distribution of studies around the summary random effects estimate (Figure 3). Egger's test confirmed the absence of funnel plot asymmetry ($t = -1.06$, $df = 10$, $p = 0.31$), indicating low risk of publication bias.

When considering the efficacy of manual therapy for pain in SIJPS, we downgraded the certainty of evidence by three levels to 'very low' due to risk of bias chiefly attributed to inadequate blinding in multiple studies (−1 level), imprecision indicated by the effect estimate crossing the null (−1 level), and inconsistency in the form of high heterogeneity (−1). We did not downgrade for the risk of publication bias, which was low.

SIJ manual therapy for disability

Our meta-regression including eight trials with nine intervention-control comparisons indicated there was a statistically significant, moderate effect of manual therapy on reducing disability in SIJPS (SMD: -0.67 ; 95% CI: -1.32 ; -0.03 , $p = 0.0418$) (Figure 4). The wide confidence interval suggests some uncertainty around the effect size estimate. Significant differences were evident between subgroups ($p < 0.01$), suggesting that the efficacy of manual therapy for disability may depend on the technique(s) used. We could not reliably compare effects on disability between subgroups, as most subgroups (three out of five) had only a single study. However, in the two subgroups with multiple studies (SMT and MET), the effect estimates had substantial overlap, indicating a plausibly similar efficacy.

High heterogeneity was present ($I^2 = 80\%$; $p < 0.01$), suggesting differences in study methods and populations may have contributed to variability in effects. We did not assess publication bias for our meta-regression for disability, given the

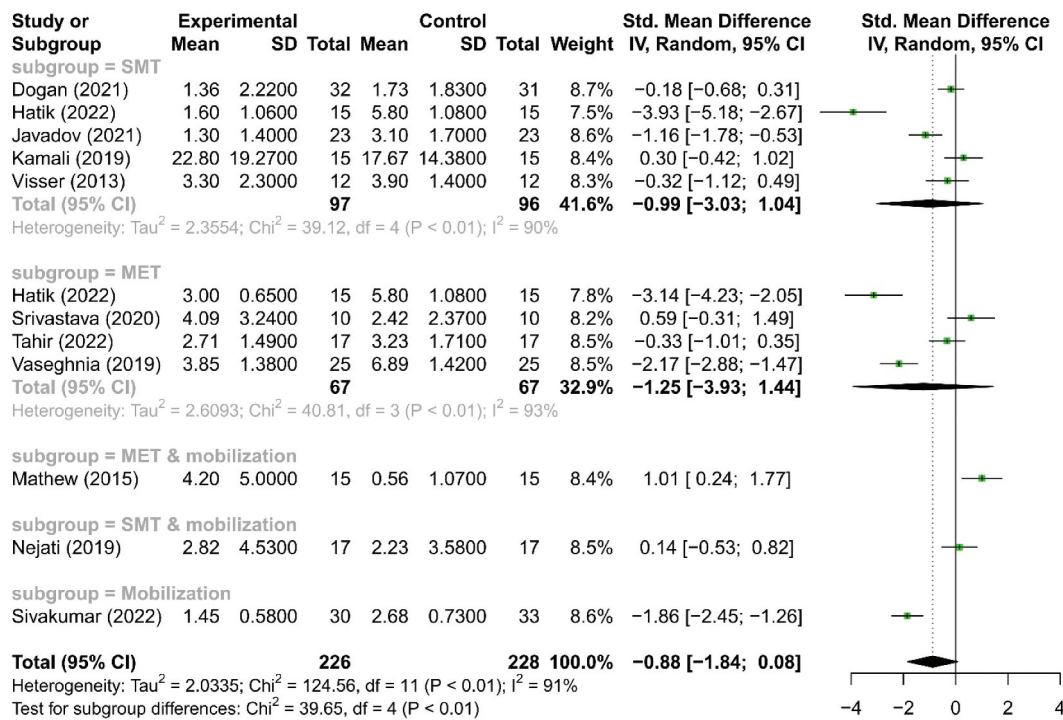


Figure 2. Meta-regression forest plot examining efficacy of sacroiliac joint manual therapy techniques on pain. Subgroups represent specific manual therapy types, each having its own effect estimate (black diamond). The bottom row (total) shows the overall meta-regression standardized mean difference for manual therapy across all studies (dotted line). An SMD of 0 indicates a null effect. A negative SMD point estimate suggests a potential reduction in pain (favors sacroiliac joint manual therapy). Mean pain scores and standard deviations are indicated in the experimental and control groups. Abbreviations: confidence interval (CI), inverse variance (IV), muscle energy technique (MET), spinal manipulative therapy (SMT), standard (std.).

consideration that Egger's test is unreliable when analyzing fewer than 10 studies [29,30]. In addition, there was overlap between the studies included in the two meta-regression models (pain and disability), making them potentially redundant with the earlier publication bias assessment for pain outcomes.

When considering the efficacy of manual therapy for disability in SIJPS, we downgraded the certainty of evidence by two levels to 'low' due to the risk of bias (−1 level) and inconsistency in the form of high heterogeneity (−1 level).

Other comparisons

Five RCTs ($n = 230$ patients) comparing different manual therapy techniques for SIJPS could not be included in the meta-analysis due to their head-to-head design without a control group [12,14,34,36,38]. In addition, each RCT examined a unique combination of manual therapies, thus failing to reach our threshold of three similarly designed studies for meta-analysis. Three of these trials noted no significant between-group difference when comparing HVLA versus myofascial release [14], MET and exercise versus MET and other exercises [38], or HVLA manipulation versus MET [34]. The other two trials found superior efficacy of MET over strain-counterstrain for disability [12] and combined SIJ &

lumbar HVLA manipulation versus SIJ manipulation alone for pain and disability [36]. Adding to the context of our meta-analysis where meta-regression also showed no clear superiority among manual therapy subgroups, these head-to-head trials also suggest uncertain comparative efficacy of manual therapy approaches for SIJPS.

Discussion

To our knowledge, this is the first systematic review with meta-analysis examining the efficacy of manual therapy on pain and disability related to SIJPS, which was strengthened by having an a priori registered protocol and multidisciplinary research team. We found very low certainty evidence that manual therapy does not significantly reduce pain and low certainty evidence that manual therapy provides a moderate and statistically significant reduction in disability among adults with SIJPS. Despite our use of meta-regression, comparison of subgroups was hindered by a small number of trials per subgroup, wide confidence intervals for effect estimates and a handful of studies that could not be meta-analyzed.

One primary outcome from our meta-analysis was that manual therapy did not significantly reduce pain among adults with SIJPS, despite having a large effect size (SMD −0.88). We offer two potential explanations.

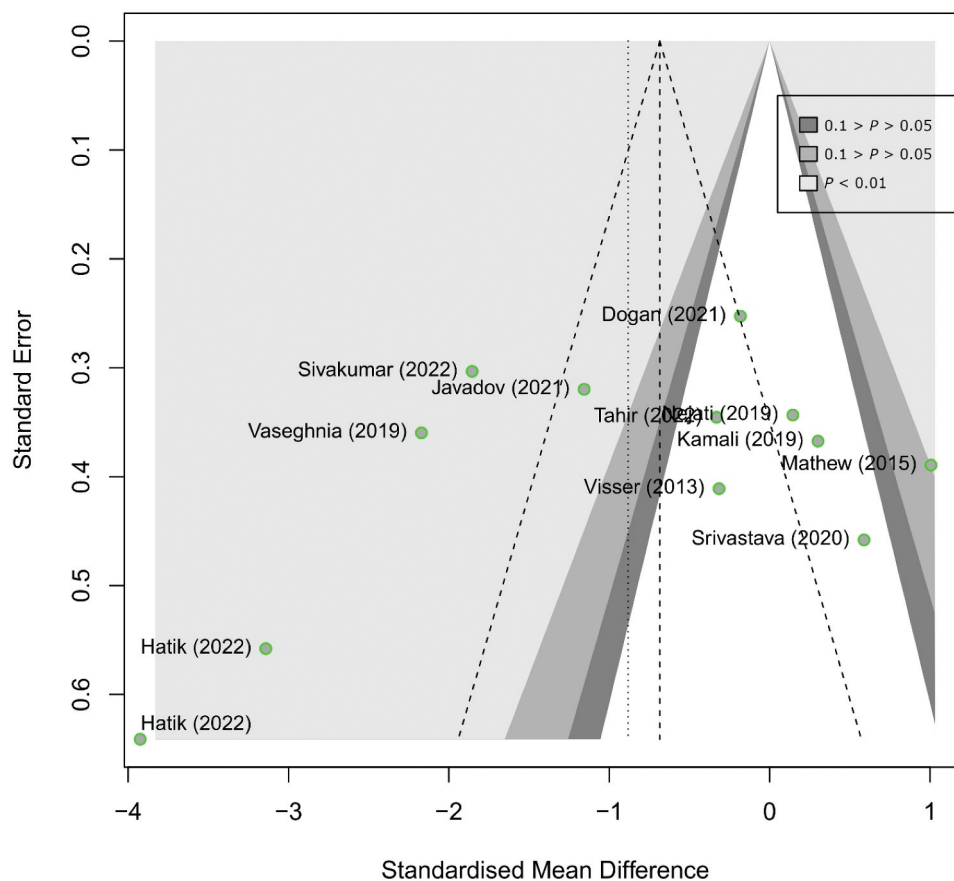


Figure 3. Contour-enhanced funnel plot. The dashed lines represent the fixed effect estimate (vertical dashed line) and corresponding 95% confidence interval limits (diagonal dashed lines), while the fainter vertical dotted line represents the random effects estimate used for our analysis. Shaded areas indicate significant p -values while the white central area indicates non-significant p -values.

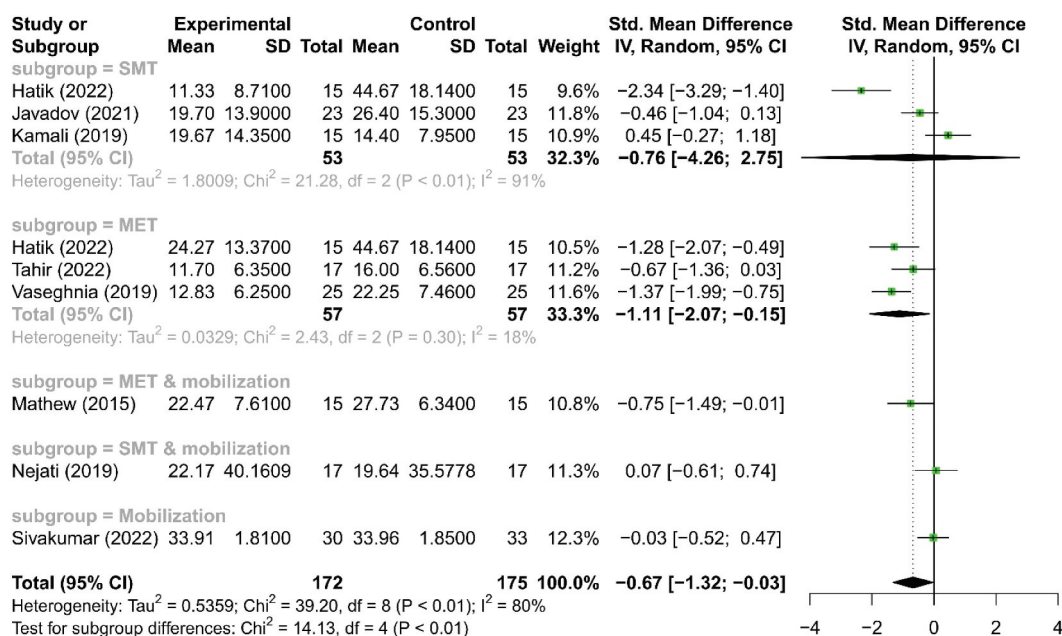


Figure 4. Meta-regression forest plot examining efficacy of sacroiliac joint manual therapy techniques on disability. Subgroups represent specific manual therapy types, each having its own effect estimate (black diamond). The bottom row (total) shows the overall meta-regression standardized mean difference for manual therapy across all studies (dotted line). An SMD of 0 indicates a null effect. A negative SMD point estimate suggests a potential reduction in disability (favors sacroiliac joint manual therapy). Mean pain scores and standard deviations are indicated in the experimental and control groups. Abbreviations: confidence interval (CI), inverse variance (IV), muscle energy technique (MET), spinal manipulative therapy (SMT).

First, given the wide confidence intervals just crossing the null value, and limitations like risk of bias and unexplained heterogeneity, a clinically meaningful effect of manual therapy on SIJPS pain cannot be ruled out until higher-quality evidence from RCTs with adequate blinding and control groups is available. Second, the most common comparator was exercise, an established treatment for low back pain and SIJPS in some studies [2]. If exercise is efficacious for SIJPS-related pain, the magnitude of between-group differences observed in our meta-analysis of post-intervention comparisons may have been reduced. Given the 'very low' certainty evidence for pain outcomes in this meta-analysis, we cannot definitively conclude whether manual therapy or exercise is superior for reducing SIJPS-related pain in adults.

Another primary outcome from our meta-analysis was that manual therapy significantly reduced disability related to SIJPS with a moderate effect size (SMD -0.67). The reasons for the improvement in disability but not pain are unclear, yet may relate to non-clinical factors. First, while there is limited evidence on the topic of outcome measure responsiveness in SIJPS [55], it is possible that changes in disability measures are more quickly captured than changes in pain in the context of the relatively short follow-up duration of the included studies (median of 4 weeks). Second, contextual effects of manual therapy should be considered [56]. Given the biomechanical nature of manual therapy, patients may expect improvements in mobility (e.g. walking, sitting, and lifting), which are components captured in ODI/MODI but not in VAS/NPRS. Finally, the disability scales encompass activities of daily living beyond just pain, potentially increasing sensitivity.

This study adds to the current literature on the topic of conservative SIJPS treatments. To date, review articles focusing on the treatment of SIJPS may not mention manual therapy in the context of such treatments [1,3]. In contrast, our synthesis suggests that manual therapy is efficacious for reducing SIJPS-related disability. Accordingly, manual therapy should be considered and/or offered as a viable treatment option for SIJPS.

The results of this study also raise important questions regarding the management of SIJD, including surgical interventions for SIJPS. At long-term follow-up in RCTs, surgical SIJ fusion has been shown to provide superior outcomes in pain and disability in patients with chronic SIJD compared to conservative care in two RCTs [57,58]. However, neither of these trials stated whether manual therapy was provided to the control group receiving conservative SIJPS management [57,58]. While our study highlights the potential utility of manual therapy as a conservative treatment for SIJPS, it remains unclear if the interpretation of surgical trials for SIJPS would differ had studies reported or used control groups incorporating SIJ manual therapy. Additional research is needed to

examine the efficacy of surgical interventions for SIJPS compared to conservative care including manual therapy.

We encountered a handful of RCTs with a unique head-to-head design (i.e. comparing two manual therapies) which prevented us from including them in our meta-analysis [12,14,34,36,38]. While these RCTs reinforced findings from our meta-regression that there is no clear superiority of specific manual therapy techniques for SIJPS, they did not examine whether manual therapy is efficacious compared to a usual care intervention. We therefore emphasize the priority for RCTs examining manual therapy for SIJPS to have a usual care control group such as exercise to establish higher-certainty evidence on this topic. Secondarily, head-to-head comparisons or network meta-analyses could be warranted to examine which manual therapies are most efficacious.

Several limitations exist for this systematic review and meta-analysis. First, data extraction was not performed independently, which can introduce bias. As the mean follow-up time for included articles was 6 weeks, a longer-term follow-up could have led to greater insights regarding the durability of treatments over time. We were unable to reliably compare the efficacy of individual manual therapy techniques due to a limited number of studies per subgroup, wide confidence intervals of effect estimates, and a unique methodological design of five head-to-head RCT designs that precluded our ability to meta-analyze them.

In real-world settings, manual therapy may or may not be provided in isolation. The use of a multimodal approach depends on clinicians' best judgment, patient preference, and other factors. While our meta-analysis examines the effects of manual therapy for SIJPS, exploration of any synergy (or lack thereof) with exercise and/or other modalities remains outside the scope of our study. Instead, the ranking of efficacy of various treatment combinations could be further examined using a network meta-analysis [59]. Considering that an optimal multimodal approach for SIJPS is currently unclear, clinicians should use their best judgment if adding manual therapy to SIJPS treatment on a case-by-case basis.

Conclusion

In this meta-analysis including 16 RCTs, we found very low certainty evidence that SIJ manual therapy did not significantly reduce pain, and low certainty evidence that SIJ manual therapy produced a statistically significant, moderate effect in reducing disability in adults with SIJPS. Subgroup analyses were limited due to low sample size, imprecise effect estimates, and inability to include some RCTs in meta-analysis due to their design, thereby precluding us from determining whether any certain manual therapy technique is more beneficial for SIJPS. While manual therapy appears to be a viable conservative

treatment option for SIJPS, further high-quality RCTs are needed to increase the certainty of evidence on this topic.

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